

ECV 101 MATERIAL SCIENCE- CAT 1

1. a.) Both K (Atm nr 19) and Ca (Atm nr. 20) are metals, Why is the melt temperature of Ca higher than the melt temp of K.
b.) The repulsion force between Na^+ ($r=0.095$) and Cl^- ($r=0.181$) ions at equilibrium is -3.02×10^{-9} N. calculate the value of constant b , given $n=9$.
(5 marks)
(5 marks)
2. a.) Define crystal structure -
b.) calculate the atomic packing factor for the BCC unit cell, assuming the atoms are hard spheres
(4 marks)
 $4r = \sqrt{2}a$
3. With aid of a diagram Explain the following terms
(6 marks)
a) Electron density
(10 marks)
b) Orbital
c) Boundary surface representation
d) Radial probability

ECV 101: INTRODUCTION TO MATERIAL SCIENCE, CAT 2

- Qn1. a). How does green concrete/cement differ from ordinary portland cement (4marks)
- b). What are the main advantages of glass-fiber-reinforced plastics (6 marks)
- strong ³ shift. - flexible
- low density - ductile
2. For a BCC unit cell (10 marks)
- a). how many atoms are there inside the unit cell 2.
- b). what is the relationship between the length of the side a of the BCC cell and the radius of its atoms $4r = \sqrt{3}a$
- c). Calculate the atomic packing factor 0.68.
3. a). With examples Differentiate between an alloy and a composite material (4 marks)
- b). Differentiate between a low carbon and high carbon steel alloys (6 marks)

09

2

3

ECV 101 MATERIAL SCIENCE- CAT1 make up

cast.
throughout.
shape of product.
sheet shaped.
finished method.
rolled.

method of production.

- a.) Metals have dense crystal structures, list the reasons for this dense packing (2 marks)
- b.) What four considerations (with 2 examples for each consideration) can classification be made Within each group of metal alloy? (8 marks)

- a.) Differentiate between thermosets and thermoplastics with examples for each (4 marks)
- b.) calculate the atomic packing factor for the FCC unit cell, assuming the atoms are hard spheres (6 marks)

3. a.) " T_{room} data cannot be used in design calculation if the part will be used at other temp" why?
- b.) With aid of a diagram, show the different methods of reinforcing plastics in a composite matrix (3marks)
- b).Using Magnesium ($Mg=12$) and Oxygen ($O=8$), explain the properties of ionic bonding (5 marks)

60

60

09

3

3

3

ECV 101: INTRODUCTION TO MATERIAL SCIENCE, CAT 1

QUESTION 1

With the aid of a schematic, show that different classes of materials have varying strength properties at varying temperature range

QUESTION 2

(10 marks)

a). Identify the key properties of a polymeric material

(4 marks)

b). Given the following information, Load applied to specimen = 69000 N, Initial specimen diameter = 1.27 cm, diameter of specimen under load = 1.2 cm. Calculate, (i). true stress (ii). true strain for the tensile test

(6 marks)

QUESTION 3

Briefly explain the criteria (checklist) for selecting the best material in engineering applications (10 marks)

Qn1. a. Future of material science is in nanomaterials, what are they? (4 marks)
Chemical substances Manufactured

b. What are superalloys and What are superalloys used for? (6 marks)
Excellent material for making

Qn2. a. Concrete solidifies and hardens after mixing with water and placement due to a chemical process known as hydration, using illustration explain how this process happens (5 marks)

b. State any five processes used in the manufacturing of plastic composites (10 marks)

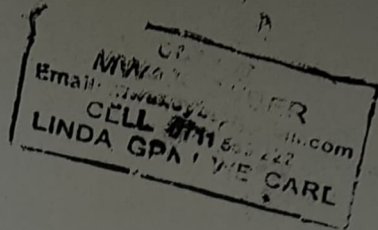
Qn3 the following creep data was obtained from a titanium alloy at 345mpa and 400oc. Plot the creep strain vs time, and determine the steady state creep rate for the test conditions

Strain, mm/mm	Time, hrs	Strain, mm/mm	Time, hrs
0.010x10 ⁻²	2	0.075x10 ⁻²	80
0.03x10 ⁻²	18	0.090x10 ⁻²	120
0.05x10 ⁻²	40	0.11x10 ⁻²	160

materials
surface to volume ratio.
high porosity
Better physicochemical properties.



KENYATTA UNIVERSITY



UNIVERSITY EXAMINATIONS 2016/2017

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE IN CIVIL ENGINEERING

ECV 101: INTRODUCTION TO MATERIALS SCIENCE

DATE: Tuesday 13th December 2016

TIME: 11.00a.m-1.00p.m

INSTRUCTIONS:

Attempt question 1 and any other two questions

QUESTION 1

1.a). (i) Define nanomaterials

(1 marks)

(ii) State the proposed advantages of using nanomaterials over their conventional counterparts

(4 marks)

b). Ductility is extent of plastic deformation that the material undergoes before fracture, describe the two measures of ductility

- Elongation
- Reduction of area

(10 marks)

c). i) Pure aluminum is a ductile metal with low tensile strength and hardness. its oxide Al_2O_3 (alumina) is extremely strong, hard and brittle. Explain this difference from an atomic bonding point of view

(5 marks)

ii) A tensile specimen of aluminium alloy is tested to fracture. At the fracture point, it has an engineering stress of 180 MPa and engineering strain 34%. Calculate the true stress and true strain at fracture

(10 marks)

QUESTION 2

2. a) Describe the terms

i) Metallic bonds

(1 marks)

(1 marks)

Phase diagram - Diagram showing phases present under equilibrium condition & phase composition at each corner of temp & overall composition

INVOLENT EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

iii) Metalloids

(1 marks)

iv) Nucleus charge effect

(1 marks)

b) With aid of a schematic, describe the phases in creep -failure test

(6 marks)

c) Discuss the checklist for selection of materials used in engineering applications

(10 marks)

QUESTION 3

3. a) Differentiate between the following mechanical terms in materials

(5 marks)

i) Cold working and Hot working

ii) Hammer and Press forging

(5 marks)

b) Discuss the major factors that affect the fatigue strength of a metal other than the chemical composition of the metal itself

(10 marks)

QUESTION 4

4. a) Define super plasticity and list the conditions under which super plasticity can be achieved?

(4 marks)

b) Differentiate between the tensional, torsional and compressive strengths

(6 marks)

c) Compare the three main methods used in hardness testing

(10 marks)

QUESTION 5

5. a) i) Differentiate between thermoplastic and thermosets

(3 marks)

ii) What are the advantages and disadvantages of injection molding

(7 marks)

b) With the aid of illustrations, describe the stages in the formation of a cup and cone ductile fracture

(10 marks)



KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2019/2020 FIRST SEMESTER EXAMINATION
FOR THE DEGREE OF BACHELOR OF SCIENCE (CIVIL ENGINEERING)

ECV 101: INTRODUCTION TO MATERIALS SCIENCE

DATE: TUESDAY, 17TH DECEMBER, 2019

TIME: 8.00 A.M. - 10.00 A.M.

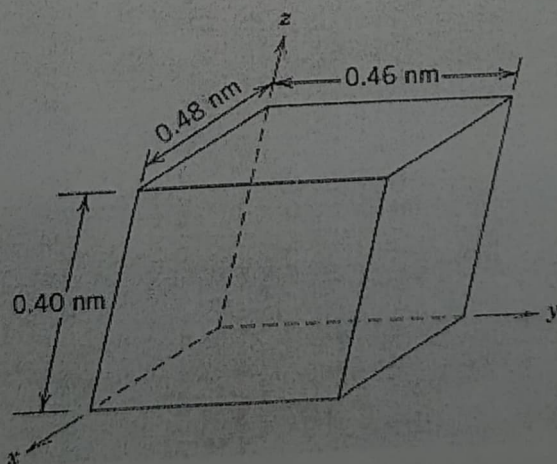
INSTRUCTIONS: Attempt Question One and any Two Questions

Question One

1. (a) (i) Metals have dense crystal structures, list the reasons for this dense packing. (2 marks)
- (ii) With aid of a diagram, show the different methods of reinforcing plastics in a composite matrix (3 marks)
- (iii) Using Sodium (Na=12) and Chloride (Cl=8), explain the properties of ionic bonding. (5 marks)
- (b) With aid of appropriate diagram (s), explain the following terms as used in atomic theory
- (i) Electron density (2.5 marks)
- (ii) Orbital (2.5 marks)
- (iii) Boundary surface representation (2.5 marks)
- (iv) Radical probability (2.5 marks)
- (c) (i) Identity the key Thermal Properties of Engineering materials. (2 marks)
- (ii) Hardness is not a fundamental property because indentation depends on shape of indenter and load applied. Show the shape of indentation (both side and top views) caused by the four main indenters types used in Hardness testing. (8 marks)

Hardness test method	Indenter	Indentation shape	
		Side view	Top view

- (9)
2. (a) (i) Both K (Atm nr 19) and Ca (Atm nr. 20) are metals, using the principle electron configuration explain why the melt temperature of Ca higher than the melt temp of K. (5 marks)
- (ii) The repulsion force between Na^+ ($r = 0.95 \text{ nm}$) and Cl^- ($r = 0.181 \text{ nm}$) ions at equilibrium is $-3.02 \times 10^{-9} \text{ N}$. Calculate the value of constant b , given $n = 9$. (5 marks)
- (b) (i) Explain how you would perform a brittleness test on a material. (5 marks)
- (ii) Using illustration, describe the process to test hardness of plastics, rubbers and other soft materials. (5 marks)
3. (a) Determine the electron configurations for the elements given below and state it is an inert gas, a halogen, an alkali metal, an alkaline earth metal, or a transition metal. Justify your choices. (6 marks)
4. (a) "Covalently bonded materials are generally less dense than ionically or metallically bonded ones" - explain this statement with relevant examples. (6 marks)
- (b) Describe with appropriate examples the key differences between Galvanic cell and oxidation. (6 marks)
- (c) For the unit cell shown in the sketch below, locate the point having coordinates $1/4 \ 1 \ 1/2$.



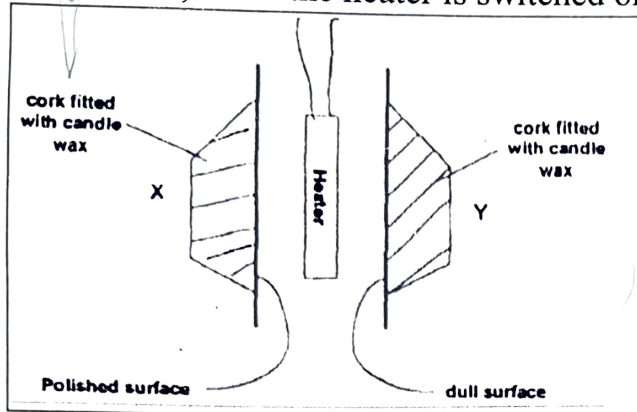
5. (a) Using appropriate schematic diagram, Illustrate the phenomena barreling in material testing. (5 marks)

(b) Using examples, discuss the main difference between crystal structure and crystal system? (5 marks)

(c) Compare Low-carbon steels to High carbon steels in terms of the checklist for Selecting the best material. (10 marks)

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1a). The figure shows two corks X and Y fixed on different plates with candlewax. Explain the observation, when the heater is switched on for a short time



i). a polished plate and a dark plate (2marks)

ii). rough plate and smooth plate (2marks)

b). Identify the differences between Brinell, Vickers and Knoop hardness tests? (6 marks)

2. Compare the engineering stress and strain with the true stress and strain for the tensile test of a low carbon steel that has the following test values. (10 marks)

Load applied to specimen = 69000 N, Initial specimen diameter = 1.27cm, diameter of specimen under load = 1.2cm

3.a). What the importance of electron configuration in bonding (2 marks)

b. With the aid of a schematic, show the effect on temperature on material properties (4 marks)

c. state the criteria (checklist)for selecting the best material in engineering applications (4 marks)

ECV101: INTRODUCTION TO MATERIAL SCIENCE CAT 2

- 1a). Concrete solidifies and hardens after mixing with water and placement due to a chemical process known as hydration, using illustration(s) explain how this process happens(5marks)
- b). Fill in the given table for the polymer composites manufacturing Methods/processes (6 marks)

Method	Advantages	Disadvantages
Injection molding		
Spray-up		
Hand lay-up		
Filament winding		
Pultrusion		

2. Differentiate between the different metal materials product processes (10 marks)

process	2 Key distinguishing properties
Cast	
wrought	
Annealed	
Drop forged	
Press forged	

3. For the following categories of properties of materials, pick a property and describe one (1) method to test the property (10 marks)

category	property	Briefly describe Method to test
physical		
optical		
electrical		
chemical		
sound		



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2020/2021

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (CIVIL ENGINEERING)

ECV 101: INTRODUCTION TO MATERIAL SCIENCE

DATE: Monday, 8th February 2021

TIME: 11.00 a.m. - 1.00 p.m.

INSTRUCTIONS:

ATTEMPT QUESTION 1 AND ANY OTHER TWO QUESTIONS

QUESTION 1

- (a) Identify a checklist Selecting the best material in material science. (4 marks) ✓
- (b) Materials are probably more deep-seated in our culture than most of us realize. List the primary classifications of solid materials, and then cite two (2) distinctive chemical feature of each. (6 marks) ✓
- (c) (i) One common item that presents some interesting material property requirements is the container for carbonated beverages. Mention 6 constraints the material used for this application must satisfy? (4 marks) ✓
- (ii) Three material (aluminum, glass, polyester, plastic) are used for carbonated beverage containers. Complete the table below showing the 2 advantages (pros) and 2 disadvantages (cons) of using each of the possible materials in packaging carbonated drinks. (4 marks) ✓

Material	Advantages (pros)	Disadvantages (cons)
Eg Glass	May be recycled	fragile

- (d) (i) Covalently bonded materials are generally less dense than ionically or metallically bonded ones. Explain why? (3 marks)

- (ii) Calculate the coulombic force of attraction between Ca^{2+} and O^{2-} in CaO , Given electron charge $e = 1.602 \times 10^{-19} \text{C}$, and the permittivity of space, $\epsilon = 8.85 \times 10^{-12} \text{ F/m}$, Ionic radius of Ca and Oxygen are 0.100nm and 0.140nm respectively. (5 marks)
- (iii) Calculate the percentage ionic character of a bond between elements A and B, given $X_A = 1.2$ and $X_B = 3.5$ (5 marks)

QUESTION 2

- (a) Some products are designed with failure essential for their function, explain three situations in which the possibility of failure is part of the design of a component or product. (6 marks)
- (b) The Stress-Strain Curves show the behaviour of a material under tensile test. Construct a Stress-Strain Curve and identify (label) all the key parts. (6 marks)
- (c) Fatigue life is influenced greatly by the method of preparation to the surfaces of the part or specimen, using illustration describe how to improve Fatigue Strength in materials. (8 marks)

QUESTION 3

- (a) Mention two reasons why are the atomic weights of the elements generally not integers? (2 marks)
- (b) Fatigue Failure is propagated through the material until critical crack is reached and material fractures, describe the S-N Curves. (8 marks)
- (c) Failure is one of the most important aspects of material behavior. It directly influences the selection of a material for a particular application, the methods of manufacturing, and the service life of the component. Describe how Plastic deformation proceeds failure in Ductile Fracture. (10 marks)

QUESTION 4

- (a) Given the atomic number for Iron=²⁶~~6~~ and sulfur=16. Give electron configurations for the elemental (Fe and S) and ionic forms (Fe^{3+} and S^{2-}). (4 marks)
- (b) Melting Point is an important consideration when the material will be subject to an operating temperature or a thermal cycle during some manufacturing processes. Describe three (3) such processes. (6 marks)
- (c) Distinguish between High carbon steels and Stainless steels. (10 marks)

QUESTION 5

- (a) Concrete is a composite material, list the key ingredients in concrete. (2 marks)
- (b) An understanding of many of the physical properties of materials is predicated on a knowledge of the interatomic forces that bind the atoms together. Complete the table which summarizes on bonding in materials below. (8 marks)

Bond Type	Bond Energy	Direction

- (c) Composites are solid materials composed of more than one substance in more than one phase. Describe the thermoplastic processes using Very short fibers for making composites.